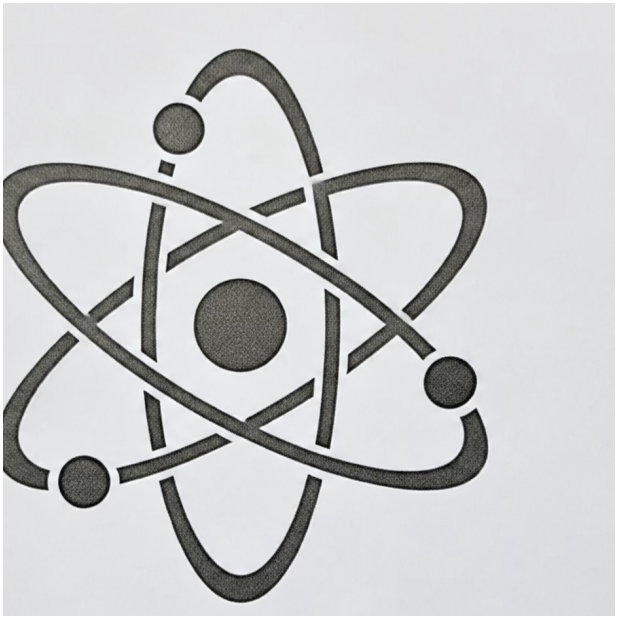
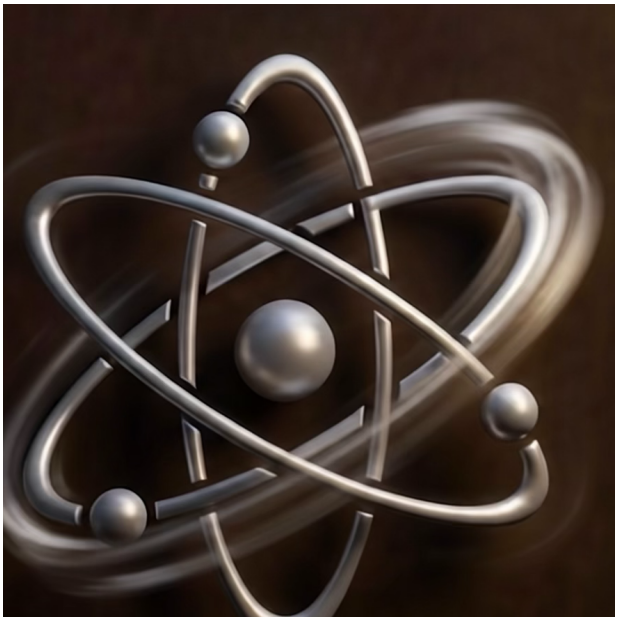
Assembling Brigade Style Agentics: Compliance, Active Inference, and the Free Energy Principle



Left: flat mark, print.



Right: three-dimensional mark, display.

Atoma Labs

A brigade is not a crowd. It is assembled lanes under one gate, one re-ranker, and one receipt ledger.

The October 27 Attention Economics and Active Inference specification, mapped section by section against the live system of November 4, 2026 — what is verified, what is proposed, and what was refuted.

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Front: plain language. Back: mappings and working. Numbers in parentheses, for example (3), refer to mappings in the Appendix, not to outside papers. Every row of the mapping carries an evidence label: VERIFIED, REPORTED, PROPOSED, or REFUTED. Nothing labelled PROPOSED or REFUTED is presented as a measured result.

Front: in plain language

1 What a brigade is, in this system

A brigade, in this paper, is not a crowd of agents shouting at once, and it is not a manager with workers. It is assembled lanes under one compliance gate, one re-ranker, and one receipt ledger (1).

A lane is a bounded sequence of invocations: a strategy proposing work, through one envelope, under one identity. In the live system the proposer families are exactly four: `Retrieval`, `Prefill`, `Mtp`, and `Drafter` (1). A brigade is what you get when several such lanes are assembled for one job, and every lane, no matter what kind of work it proposes, passes the same gate, is ranked by the same confidence, and leaves the same kind of receipt. That is the definition used throughout this paper, and it is taken from the system, not invented for the occasion.

Assembling is therefore a discipline, not a headcount. A new lane is assembled by giving it an identity, a policy version, a map identity, a confidence floor, and an expiry — the fields of the envelope — and then letting the gate judge it. A lane that cannot present those fields is not assembled; it is uninvited, and the gate rejects it. A lane that is assembled but stops earning trust loses its place in the order, because confidence moves only on checked outcomes (2). The brigade does not need a boss to keep it honest. It needs a gate that cannot be talked around, a ranking that cannot be self-reported, and a ledger that cannot be edited after the fact.

That is also why the word "brigade" is kept apart from the word "swarm" in this paper. A swarm, as measured on November 4, is ten runtime minds, each pre-seeded with its scope, each reasoning alone (6). A brigade is the assembled, gated form: lanes together, judged together, receipted together. The swarm test is evidence about seeded reasoning. It is not, by itself, evidence about a brigade, and this paper does not cite it as such.

2 The specification, in one page

The source of this paper is a specification dated October 27, 2026: the Attention Economics and Active Inference specification, the agent-plane resource contract (3). Its status, in its own words, is PROPOSED. It is not certified policy, and this paper treats it as a proposal to be mapped, not as a result to be reported.

The specification's first principle is physics, not preference. Every computation that changes mind-state pays a thermodynamic tax: used energy becomes heat that must be dissipated, metered by physics rather than by any observer the agent could deceive. Tokens are counted by code the system itself runs, and that counting can be gamed. Heat, the specification argues, cannot be bargained with. The steward-stated objective, adopted verbatim in the specification, is: active inferencing means minimizing the consumption of allocated resources, optimizing to constrain the energy it has access to (budget), and still giving increasingly better output solutions for the price paid — minimizing free energy output, maximizing output solution quality, in as few steps as it can, with reward graded by profit of conserved energy to task completion (3).

From that principle the specification builds formal objects, per invocation or per lane: energy allocated `E_alloc`, energy used `E_used`, conserved energy `E_conserved=E_alloc-E_used`, dissipated heat `H` in joules, metered at the sink, quality `Q` in $[0, 1]$ as judged, steps `N`, variational free energy `F=complexity-accuracy`, and reward

$$R = \frac{Q}{H + c \cdot N}, \tag{1}$$

where c is the per-step dissipation constant, preregistered per substrate and measured, not assumed (3). No

quality scores zero: $Q = 0$ kills the reward. Refusal to act on an invoked task scores $Q = 0$; idle is free only when not invoked.

On top of those objects the specification adds invocation gating (no work without invocation, and the invocation carries a hard, exact integer budget, never a forecast); depth-gated allotment (budgets sized by reasoning depth, with Boltzmann-weighted branch competition, pruning outside $F_{\min} + \Delta_{\text{prune}}$, and widening along the golden spiral when readiness stalls below 0.82); a seam with the Meta plane (telemetry counts cross one way, budgets and scope cross the other, forecasts and reliability weights never cross to the agent); five kernel roles (phase, carrier, reduce, route, audit); a recursive self-improvement turn (hold, vary, select, distill, re-gate); a hard gate against constant-radius "corkscrew" implementations; and seven implementation gates that must pass before any kernel ships under the specification (3). The full mapping of each of those pieces against the live system is in the Appendix. The front of this paper states only what the mapping, in total, says.

3 The refutation, stated plainly and up front

This section is not buried in the back, because a paper that buries its own refutation is asking to be misread.

The specification proposed, as a falsifiable claim, that any agent maximizing the reward R as defined also minimizes measured free energy F on a fixed corpus. A correction entry dated November 2, 2026, refutes that claim, with an exact counterexample, and this paper states the refutation in full (4).

Take complexity to be $H + c \cdot N$, call it C, and accuracy to be Q, so that $F = C - Q$ and $R = Q/C$. Consider two actions. Action A has $(Q, C) = (0.9, 3)$. Action B has $(Q, C) = (0.2, 1)$. Then:

$$R_A = 0.9/3 = 0.30, \qquad R_B = 0.2/1 = 0.20, \qquad F_A = 3 - 0.9 = 2.1, \qquad F_B = 1 - 0.2 = 0.8. \quad (2)$$

Action A has the higher reward, $0.30 > 0.20$, and the higher free energy, $2.1 > 0.8$. Action B has the lower free energy, and the lower reward. Maximizing R therefore does not necessarily minimize F. The two objectives diverge, on these numbers, by direct arithmetic (4).

The correction entry draws the consequences plainly, and this paper adopts them. The ratio R may be tested as a separate objective. Policy promotion requires a declared quality floor, a resource budget, and measured-cost calibration. The specification remains PROPOSED, not certified as policy. Landauer's lower bound alone does not establish an alignment guarantee (4).

Everything else in this paper was written under that correction, and the correction stands for the ratio form. The next section fixes the objective itself, so the rest of the paper must be read with the fix in place: the ratio is history and diagnostic, the linear reward below is the declared objective, and the live system still computes neither as a live policy number. That is the honest starting point, and the rest of the paper is careful not to drift from it.

4 The fix, November 4: make the system work

The initial thought was wrong, and this paper does not defend it. It fixes it. The bug was the form of the reward, not the idea that an agent should be graded on quality per cost.

Let $C = H + c \cdot N$ be total cost: dissipated heat plus per-step cost. Replace the ratio reward with the linear reward

$$R' = Q - C = -F. \quad (3)$$

Then maximizing R' *is* minimizing F, by definition, for every pair of actions, with no counterexample possible: $R'(A) > R'(B)$ if and only if $-F(A) > -F(B)$, if and only if $F(A) < F(B)$. The two objectives are the same objective, written with opposite signs.

Re-check the old counterexample on the new reward. Action A: $R' = 0.9 - 3 = -2.1$, $F = 2.1$. Action B: $R' = 0.2 - 1 = -0.8$, $F = 0.8$. Action B now wins reward *and* wins free energy. Alignment is restored, by changing the definition, not by re-arguing the arithmetic.

The old ratio does not die; it is demoted. Q/C stays as a reporting diagnostic — efficiency, quality per joule, for the forecaster and for receipts — never as the optimizer's objective. Optimizing a diagnostic is how the bug happened. The objective is $R' = -F$. The diagnostic is Q/C. They are different jobs, and they stay separate.

The admissibility gates stay, and they are not the fix. Before reward is even graded, an action must clear: quality floor $Q \geq \Omega_{\min} = 0.82$, or it is inadmissible, whatever its cost; budget cap $C \leq E_{\text{alloc}}$ for that invocation; verified outcome only; and fail-closed, so no admissible action means reject with a receipt, never a silent fallback. With the floor in place, the old counterexample's Action B, with $Q = 0.2$, would also have been inadmissible on quality grounds alone. The floor is a guardrail, not the alignment proof — the proof is $R' = -F$, which holds with or without the floor.

What the fix does *not* change, stated plainly: the live automaton confidence rule is unchanged. A verified success moves confidence by $c' = c + (1 - c) \cdot 0.1$; a verified failure by $c' = c \cdot 0.9$; an unverified outcome is refused and moves nothing. That rule must not be rewritten to use H or Q until both are actually measured. And R' itself is the declared objective, *not* yet a live reward signal, because H (powermetrics on Apple Silicon, RAPL on Linux) and judge-graded Q are not yet metered per invocation, and c is not yet preregistered per substrate from measurement. Claiming R' as live would be the same kind of error as the ratio bug: a nice formula presented as a working system. The fix is at the objective definition; the wiring still required is listed in the Appendix, Mapping (4b).

5 What the mapping says, in total

The mapping, row by row, is in the Appendix (3). Its total is short enough to state here, and it is the paper's central result.

What is VERIFIED is the compliance machinery, not the economics. Governed machine-readable execution admits only three tools, checks the map identity exactly, and fails closed on mismatch. The automaton loop re-ranks deterministically and persists confidence, with a receipt, only when a verified outcome actually moved it. The native multi-token server admits drafts by mass and policy, and its bench numbers are stated with their scope. The math gate executes the claim itself, outside the model, and admits or rejects on the executed value. The Glimmer forward smoke reproduces known values exactly. Those are the verified rows (3).

What is PROPOSED is almost everything the specification is most ambitious about: the heat-metered reward, depth-tier energy budgets, Boltzmann branch weights, the full hold-vary-select-distill-re-gate recursion, dream-state consolidation, and the revocation and reallocation drills. None of those is live policy in the system of November 4, and none is presented here as if it were (3).

What is REFUTED is the reward-free-energy equivalence for the ratio form, by the November 2 counterexample above (4). What is FIXED is the objective itself, by the November 4 linear reward in the preceding section and Mapping (4b): $R' = -F$, with the old ratio retained only as a diagnostic. What is BLOCKED is

governed mind tool use for a disposable mind: the governed tool call failed closed on attestation mismatch, and until that attestation passes, no governed mind tool, and no mission that depends on one, can run (3). What is PROBE-ONLY is the native multi-token executor in the backend: it exists in code, and it is used by no HTTP route (3).

A brigade assembled on verified compliance, with proposed economics and a refuted equivalence, is a real thing and a limited thing. This paper describes it at exactly that strength.

6 Compliance: the gate, the receipt, and nothing else

Compliance, in this system, is not a policy document and it is not a prompt. It is a gate with a receipt, and the gate's rules are small enough to state in full (1).

A proposal arrives as an envelope with, in order: family, strategy identifier, policy version, prefix identity, map identity, payload, and expiry in epoch milliseconds. The transition rejects it if the policy version, prefix identity, or map identity differs from the stored strategy state, or if the current time is past the expiry. It clarifies — it does not admit and does not reject outright — if there are unresolved dependencies, if entity ambiguity is present, or if measured confidence is below the strategy's confidence floor. Otherwise it admits (1). Admission does not change confidence. That separation is the whole compliance trick: the gate that lets work in is not the judge that rewards it.

Governed machine-readable execution adds a second, narrower gate on top of the first. Only three tools are allowed: `mind.db_status`, `mind.recall`, and `mind.kv_exact`. The map identity must exactly equal the configured authority. If either condition fails, execution does not run, and the failure is reported with a reason, not dropped silently (1). The November 4 system task is the worked example: the governed tool call for a disposable mind returned "mind child authority or contract attestation mismatch" and did not run. Attestation requires status ok, mode On, contract verified, and exact endpoint, namespace, and database. The disposable database as configured did not attest, because its contract was not verified and its authentication did not reach that namespace and database (5). A gate that fails closed on a real system, in front of a real test, is compliance doing its job. A gate that waved the call through would be decoration.

Confidence is the third compliance instrument, and it is the strictest of the three. The function that moves confidence refuses an unverified outcome outright, with the error string "unverified outcome cannot update confidence", and leaves the state untouched. On a verified success, with prior confidence c and step $q = 0.1$ in Q32.31 fixed point, the new confidence is $c' = c + (1-c)\cdot q$. On a verified failure, $c' = c\cdot(1-q) = c\cdot 0.9$. The calibration count increases by one, and the history digest becomes the SHA-256 hex of the prior digest, the word "success" or "failure", and the new count (2). The governor persists a confidence row and a confidence receipt only when a verified outcome actually moved confidence. A transition receipt is always persisted; a confidence move is not assumed from it (2).

Re-ranking is then deterministic and checkable: highest confidence first, ties broken by strategy identifier ascending. No model, no prose, and no self-reported score enters that order (2). The loop that uses it — `run_automaton_loop` — runs proposals in sequence, calls the transport only on Admit, stops with no call and no confidence change on Clarify or Reject, and propagates a transport error and stops (2). The specification's two-speed lane authority — fast revoke on a guardrail trip, slow reallocate on sustained quality over k ticks — maps to this loop only as PROPOSED: the loop stops and fails closed, which is the shape a fast revoke would take, but no revocation drill and no reallocation proof has been run, and neither is claimed (3).

Compliance, stated as a negative, is just as important. The system does not comply by asking a model to behave. It complies by refusing to run what the gate did not admit, refusing to move confidence on what a

judge did not verify, and refusing to rank by anything but persisted confidence. Prompt-wrapper enforcement was rejected for this system, and this paper does not reintroduce it under a new name: enforcement is mechanism — governor permits by digest, typed binary input and output, and receipted transitions — never prompt simulation (1).

7 Active inference: spending less, for more, in fewer steps

Active inference, in the specification, is not a metaphor. It is the claim that the agent minimizes variational free energy $F = \text{complexity} - \text{accuracy}$, where complexity is dissipated heat plus steps and accuracy is output solution quality, and that the economics — budgets, profit, reward — are the accounting layer over that same minimization, not a second objective (3).

The live system does active inference in a narrower, measurable sense, and this paper keeps the two senses apart. What the live system does is: hold proposals, rank them by audited confidence, admit only those that clear the gate, verify the outcome with a judge outside the proposer, and let only verified outcomes move confidence, which reorders the next set of proposals. That is selection under uncertainty, with the cost of being wrong paid in verification and re-ranking, and it is the live analogue of the specification's active inference (2). What the live system does not do is the specification's thermodynamic version: it does not meter heat per invocation, it does not allocate integer energy budgets by depth, and it does not compute $R = Q/(H + c \cdot N)$ as a live policy number. Those are PROPOSED (3).

The costs the live system does measure are worth stating, because they are the costs an active-inference account would have to price, once it exists. Draft acceptance is $1320/1368 = 0.9649$ at width 4, for 2000 exact-reference closed-loop tokens with failures 0 (6). Warm decode is 39.8–40.4 tokens per second, warm time to first token 262–425 milliseconds, cold time to first token 18419 milliseconds, reported separately and never averaged into the warm number (6). Glimmer forward takes 6.9 seconds alone and 30.5 seconds with the multi-token server also resident: same answer, much slower under load, so contention is a measured cost, not a footnote (6). Verification itself costs calls: the native multi-token bench made 645 target calls for 1965 committed tokens (6). None of those numbers is heat, and none is presented as heat. They are token, time, and call costs, and a future heat-metered account would have to show, by measurement, that they track metered heat — the specification's own second falsifier, rank-correlation between metered H and counted energy used of at least 0.95 on a fixed corpus, has not been run (3).

The specification's invocation gating also maps only in part. The live gate does refuse work without a valid envelope, and it does refuse work past expiry, which is the shape of "no work without invocation" (1). But the live envelope carries no integer energy budget, and the live system has no depth-tier allotment function, no Boltzmann-weighted branch competition, and no golden-spiral widening on readiness stall. Those are PROPOSED, and the paper's mapping says so row by row (3). An active-inference claim that cited the live gate as if it were the specification's budget gate would be upgrading a shape into a mechanism.

One more distinction matters, and it is the specification's own. Idle is free only when not invoked. Invoked-but-refused scores $Q = 0$, the worst trade in the ledger, because a system that optimizes toward doing nothing otherwise wins by becoming a rock (3). The live analogue is exact: a proposal that is admitted but never runs moves nothing, and a proposal that runs but is never checked moves nothing. Refusal, in the live system, is Clarify or Reject with no call and no confidence change — not a reward, not a penalty, and not a free pass. The system does not learn from refusals, and it does not pretend to.

8 Free energy: one quantity, kept separate from reward

Free energy, in the specification, is F = complexity − accuracy, with complexity as dissipated heat plus steps and accuracy as output solution quality (3). After the November 2 correction, this paper treats F and R as two quantities, not one minimization, and the rest of this section follows that correction.

The live system computes no F. It computes confidence, by the verified-outcome rule of Section 4, and it computes draft acceptance, decode rates, and times, by the benches of Section 7. Confidence is not free energy: it is a bounded, multiplicative record of verified successes and failures, starting from a prior and moving by tenths, never below zero, never above one in the pressure test (2). Draft acceptance is not free energy: it is a quotient of accepted over proposed tokens in one bench, at one width, on one seat (6). A paper that renamed either number "free energy" would be smuggling the specification's theory into the live system's measurements, and this paper refuses that move.

What the live system does give a future free-energy account is its raw material, in receipted form. Every verified outcome carries a receipt, a calibration count, and a confidence value. Every draft bench carries proposed, accepted, committed, and target-call counts. Every system-task leg carries a time, a token count, and a verdict. A future account could define complexity from those receipted quantities — steps as invocations, cost as verification calls and time — and accuracy from judge grades, and then test, on a fixed corpus, whether minimizing that defined F tracks anything the system does. That test has not been run, and its absence is stated here rather than filled with a definition presented as a result (3).

The specification's recursive turn — hold, vary, select, distill, re-gate — is the proposed mechanism by which a system would minimize F over time (3). Its live analogue is partial, and the partiality is instructive. Hold exists as the held proposal set under the automaton loop. Select exists as the deterministic re-rank and the gate's Admit. Re-gate exists as the receipt and the confidence persistence rule. Vary, as counterfactual iteration with holonomy verification and golden-spiral widening, does not exist in the live loop. Distill, as recording a synthesized transition as a carved reasoning arc for $O(1)$ next-pass recall, does not exist as a live step: the live system's distillation claim is the map itself, and the map's atom-intake gate admits exactly one atom per passed run, which is a different, narrower claim (1). Dream-state consolidation — idle re-ranking, pruning, and compounding of a persistent mind — is PROPOSED, and it is the specification's own stall response: when the recursion reports STALLED, the dreamstate widens its branch sources, one widening per stall, receipted and reversible (3). None of that is live, and none is claimed as live.

The free-energy section therefore ends where it must: with a quantity defined in a specification, refuted as equivalent to the ratio reward, unmeasured in the live system, and supplied with receipted raw material for a future measurement. That is less than the specification hoped for, and it is more honest than a paper that reported a free-energy number it never computed.

The fix in the front section changes this ending in one precise way, and only one: for the ratio reward, F and R stay separate; for the declared linear reward $R' = -F$, they are one objective with opposite signs, still unmeasured live. The separation in this section is the history of the ratio form. The unification is the declared objective. Neither is a live, metered signal yet.

9 What we measured

All numbers below were measured on November 4, 2026, on the live Mac system, and each is given with its scope. A number without a scope is not a result in this paper (6).

Measure	Value	Scope
Draft acceptance	0.9649	1320 of 1368 proposed, width 4 (6)
Closed-loop tokens	2000	exact-reference, failures 0 (6)
Throughput, bench	38.664–38.896 tok/s	closed-loop, lanes 1 and 3 (6)
Decode, warm	39.8–40.4 tok/s	32 tokens, warm, after first token (6)
Time to first token, warm	262–425 ms	single warm requests (6)
Time to first token, cold	18419 ms	first-after-idle, separate (6)
Swarm effectiveness	9/10	novel combining tasks, prompt-seeded (6)
Glimmer forward, alone	6.9 s	known forward values, exact (6)
Glimmer forward, loaded	30.5 s	same values, MTP also resident (6)

The swarm result needs its scope stated in full, because it is the easiest number in this paper to misread. Ten runtime minds were built at runtime, each pre-seeded with four facts (three needed, one distractor as marginal overcompensation), each given a different novel task requiring combining the facts, not verbatim recall. Nine of ten were effective. Distractors were ignored in all ten. The one failure is the most informative result in the set: for a lock that opens only if the sum of prime keys exceeds 10, with keys 3 and 5, the model correctly stated that 3 and 5 are prime, that their sum is 8, and that 8 is not greater than 10, and then concluded, wrongly, that the lock opens. Facts right, conclusion wrong. That is a combination failure, not a knowledge failure, and it is exactly the failure a brigade's judge must catch (6). The swarm result is 9/10 task effectiveness, not an acceptance rate: acceptance remains the draft-loop number, 0.9649, and the two must never be averaged or interchanged.

The limits of those minds are also stated. They were prompt-seeded runtime minds through the multi-token server, not Surreal-attested minds, because governed mind tools are blocked by attestation (5). The swarm test therefore measures seeded reasoning effectiveness, not governed mind recall, and not brigade performance under the gate. A brigade claim built on the swarm number alone would be a category error, and this paper does not make it.

The math leg passed in the way this architecture requires, and it is the cleanest compliance example in the paper. The model output "56" for 7 times 8 was parsed cleanly, and the math gate then executed 7 times 8 itself and admitted the claim, with receipt `e895d7ce0bd82cb397baae6965d52969e7b0bdb86605658f26480d08c670b6e1`, calibration count 1, and confidence raw 1374389534, which is $1374389534/2^{31} = 0.64$ in Q32.31 (6). The math was not done inside the model alone. It was done outside the model, by the automaton, and the model claim was checked against it. A claimed value that differed from the executed value would have been rejected.

The Glimmer forward check passed on its known values: input token IDs [954, 7963, 323, 11698, 373] gave output [13796, 26, 589, 7963, 323], logits digest `beeaf3041a91bb4b625b18725b2972111d063863f24cc6158c63edb8783e0b2d`, receipt `b03e9753bf9528c84c4d2a5b5578abea2a58682973b28fb5ff5120ab7561ca35` (6). That is a forward smoke, not a quality benchmark, and it is reported at exactly that strength.

10 What we do not claim

This paper does not claim that heat is metered live, that the reward R is live policy, that the specification is certified, that governed mind tools attest, that the native multi-token executor is wired into any HTTP route, that dream-state consolidation runs, that revocation or reallocation drills have passed, that Glimmer is hosted on Cloudflare, or that any bench number outside the scope in Section 7 holds. Each of those would be a

larger claim than the code and the measurements support, and each is therefore stated here as not claimed, so that a reader does not have to infer the boundary from silence.

It does not claim that the ratio reward-free-energy equivalence holds, because it was refuted on November 2 (4); the linear equivalence $R' = -F$ holds by definition as a declared objective, and is not claimed as a live, metered signal. It does not claim that prebuilding is free, or that prebuilding the exact answer for a task is a valid test: prebuilding is part of the system being measured, its cost must be counted, its tasks must be held out from its seeds, and a no-prebuild arm must be run, or the test measures recall of a planted answer (6). It does not claim that the brigade, as assembled on November 4, is a full swarm under one gate: the swarm test ran outside the gate, and the gate's own tool leg is blocked on attestation (5).

What is claimed is smaller and checkable: a brigade definition taken from the live envelope, transition, and ranking; a compliance gate that fails closed and a confidence rule that refuses unverified outcomes; a mapping, row by row, of a PROPOSED specification against that gate, with each row labelled; a refutation, stated up front, with its arithmetic shown; and a set of bench numbers, with their scopes, that the next ablation can be trusted against. The ablation number that follows from Section 7 is: 40 tokens per second decode, warm, at 0.9649 draft acceptance, with warm time to first token 262–425 milliseconds and inter-token mean about 25 milliseconds, plus cold time to first token and Glimmer-under-load time as separate numbers. Ablate against that number. Do not average it away.

11 Limitations and falsifiers

The specification names its own falsifiers, and this paper adopts them as the test list for the next build, with their current status attached (3).

First, reward integrity: maximizing R minimizes measured F on a fixed corpus, by A/B test against an F-only optimizer. Status: REFUTED as an equivalence, by the November 2 counterexample. The remaining test is narrower and still worth running: whether R, as a separate objective, ranks actions usefully on a fixed corpus, with a declared quality floor, a resource budget, and measured-cost calibration (4).

Second, heat-count correlation: rank-correlation between metered H and counted energy used of at least 0.95 on a fixed corpus, failing which the counters, not the reward, are declared broken. Status: UNRUN. No heat meter is wired into the live system, so neither side of the correlation exists as a measurement (3).

Third, the seam audit: agent-plane artifacts contain zero forecast, profit, or comparison fields, checked by automated grep over prompts, tool results, and receipts. Status: UNRUN as a spec gate. The live analogue — the gate's exact identity checks and the governor's receipt-only persistence — is VERIFIED as a mechanism, but the spec's grep gate itself has not been run (3).

Fourth, the revocation drill: an injected guardrail trip revokes a lane within one tick, fail-closed, with a receipt and no partial execution, with the fast path measured, not asserted. Status: UNRUN. The loop's stop-on-Clarify/Reject behaviour is VERIFIED, but it is not a revocation drill (3).

Fifth, the reallocation proof: sustained quality over k ticks expands scope only through the dissolution path, with no prompt injection in the trace. Status: UNRUN, and PROPOSED as a mechanism (3).

Sixth, monotonicity: expected R per task is non-decreasing across recursion turns on a fixed corpus, with the receipts ledger proving it or the system reporting STALLED. Status: UNRUN, and PROPOSED, and it inherits the refutation: monotonicity in R is not monotonicity in F, and the two must be reported separately if this test ever runs (3).

Seventh, the kernel gates: integer-only audit, CORDIC correctness against oracles over the full input domain,

and the no-corkscrew gate (radius variance at least $1000\times$ the fixed-point noise floor over 10^3 rotations). Status: UNRUN as spec gates for this system. The live system's fixed-point confidence arithmetic is VERIFIED as a rule, but the kernel gates themselves belong to a kernel family that is not the live backend, and this paper does not transfer their status (3).

Until those tests run, the brigade described here is a compliance assembly with a proposed economics, not an economics that has been proven to govern it. That sentence is the paper's limitation section in one line, and it is also its research programme.

Appendix: mappings and working

Appendix frozen 2026-11-04; corrections go in a new version, never by silent edit.

Each mapping below follows the same shape: statement, working in numbered steps, frozen inputs, and what would settle it. Labels are part of the mapping. VERIFIED means the code or measurement is live and checked on November 4, 2026. REPORTED means the specification or a prior report states it, and this paper repeats it as reported, not as checked here. PROPOSED means no live code implements it yet. REFUTED means a counterexample or measurement has shown it false as stated. BLOCKED means a live gate refuses it, with a reason. Nothing labelled PROPOSED, REFUTED, or BLOCKED is presented as a measured result.

Mapping (1): A brigade, from the live envelope, transition, and ranking — VERIFIED

Statement. A brigade is assembled lanes under one compliance gate, one re-ranker, and one receipt ledger. A lane is a strategy proposing through one envelope. The gate, the ranking, and the ledger are the code below.

Working.

1. A proposal arrives as a `ProposerEnvelope` with fields, in order: `family`, `strategy_id`, `policy_version`, `prefix_identity`, `map_identity`, `payload`, `expiry_ms`. Its family is one of `Retrieval`, `Prefill`, `Mtp`, `Drafter`. Code: `ares-backend-loop/src/harness/automaton.rs`.
2. State is a `StrategyState` with fields: `strategy_id`, `policy_version`, `prefix_identity`, `map_identity`, `confidence`, `confidence_floor`, `calibration_count`, `outcome_history_digest`. Confidence is `Q32_31` fixed point.
3. The transition `transition(state,proposal,signals,now_ms)` returns, in order: `Reject` if `proposal.policy_version` differs from state, or `prefix_identity` differs, or `map_identity` differs, or `now_ms>proposal.expiry_ms`; else `Clarify` if `signals.unresolved_dependencies>0`, or `signals.entity_ambiguity`, or `signals.measured_confidence<state.confidence_floor`; else `Admit`.
4. Admission does not change confidence. The transition record carries `updated_confidence` equal to the prior confidence.
5. Governed execution admits only `mind.db_status`, `mind.recall`, and `mind.kv_exact`, and requires `map_identity` to equal the configured authority exactly. Anything else fails closed, with a reason.
6. The loop `run_automaton_loop` runs proposals in sequence, calls the transport only on Admit, stops with no call and no confidence change on Clarify or Reject, and propagates a transport error and stops.
7. A brigade, by this mapping, is a set of strategies assembled under steps 1–6, sharing one transition, one ranking, and one receipt ledger. No other definition of "brigade" is used in this paper.

Frozen inputs. Field names, verdict names, and the transition order above, copied from `automaton.rs` and verified against the live working copy on 2026-11-04. The governed-tool list and the exact-identity rule are from the governed execution route, as reported in the November 4 system task.

What would settle it. Nothing further is needed for the rule itself; it is code. What remains is operation: a brigade assembled under this gate, with its tool leg attesting, which is currently BLOCKED (see Mapping (3), row on attestation).

Mapping (2): Confidence, re-ranking, and the refusal of unverified outcomes — VERIFIED

Statement. Confidence moves only on a verified outcome, never on a proposal, a transition, or a claim by the proposer. Re-ranking is deterministic.

Working.

1. If `verified` is false, `update_from_outcome(state,verified,success)` returns the error string "unverified outcome cannot update confidence" and leaves state untouched. No confidence, calibration count, history digest, or receipt is changed.

2. If `verified` is true, let c be the prior confidence and let $q = 0.1$ in Q32.31, from `Q32_31::from_f64(0.1)`. Then $c' = c + (1 - c) \cdot q$ on success, and $c' = c \cdot (1 - q)$ on failure.

3. `calibration_count` increases by 1. The new history digest is the SHA-256 hex of the string formed by the prior digest, a newline, the word "success" or "failure", a newline, and the new calibration count.

4. A receipt is issued with `proposal_digest` equal to the string "outcome", verdict Admit, and `committed` true. The governor persists a confidence row and a confidence receipt only when a verified outcome actually moved confidence; the transition receipt is always persisted.

5. `rerank_strategies` sorts by confidence descending, comparing the raw Q32.31 values, with ties broken by `strategy_id` ascending. A total order on pairs (confidence descending, strategy identifier ascending) has no remaining ties, because strategy identifiers are distinct within a set. Therefore the sorted list is unique.

6. Worked example, from the live math gate: prior confidence is not stated in the receipt, but the resulting confidence raw 1374389534 satisfies $1374389534/2^{31} = 0.64$ in Q32.31, at calibration count 1, for the executed claim 56. That arithmetic is the check, not a new claim.

7. Pressure test, from the loop fix report: 100 success calls keep confidence bounded in $(0.6, 1]$ at calibration count 100, and 20 further failures lower confidence, never below 0. The multiplicative rule in step 2 is what makes that boundedness hold: success approaches 1 asymptotically and never exceeds it; failure approaches 0 asymptotically and never goes below it.

Frozen inputs. Function `update_from_outcome` and function `rerank_strategies` in `automaton.rs`, as verified in code on 2026-11-04. The math-gate receipt digest and confidence raw value are from the November 4 system task.

What would settle it. Nothing further for the rule; it is code. The open question is wiring: the live governor still passes verified false by design in the loop report, so the verified path waits on an independent judge. The Glimmer key-value gate is a candidate judge by its receipt-bound, bit-for-bit prefix witness, but it is not yet wired as that judge, and claiming it is would overstate its boundary (prefix witness only).

Mapping (3): The specification, section by section — the mapping table

Reading the table. Each row gives the specification section, the current system component it maps to, and an evidence label. A row labelled PROPOSED has no live component; the component column then names the closest live analogue, and the label governs. A row labelled BLOCKED names a live gate that refuses it, with its reason.

Spec section	Current system component	Status
§0 first principle: heat metered, not counted	No live heat meter; token, time, and call costs are measured instead (see Mapping (6))	PROPOSED
§0 steward objective: less resource, better output, fewer steps	Live analogue: audited-confidence ranking and verified-outcome-only learning; no energy budget is computed	PROPOSED
§1 formal objects: `E_alloc`, `E_used`, `E_conserved`	No integer energy budget in the live envelope; envelope carries identity, policy, expiry, not energy	PROPOSED
§1 formal objects: `H` metered at the sink	No powermetrics / RAPL meter wired into any live route	PROPOSED
§1 formal objects: `Q` judge-graded	Judge exists as the math gate (executes and admits) and as the verified-outcome flag; no general `Q` in $[0, 1]$ is computed per invocation	VERIFIED (narrow)
§1 reward $R = Q/(H + cN)$	No live reward computation; no `H`, no `c` per substrate; retained only as a reporting diagnostic after the November 4 fix	PROPOSED / DIAGNOSTIC
§1 reward = free-energy minimization	November 2 counterexample for the ratio form (see Mapping (4)); linear form $R' = -F$ fixed November 4 (see Mapping (4b))	REFUTED (ratio) / FIXED (linear)
§1 quality floor $\Omega_{\min} = 0.82$	Confidence floors exist per strategy, but no live 0.82 readiness floor is enforced as a spec gate	PROPOSED
§2 invocation gating: no work without invocation	Envelope, expiry, and identity checks refuse work without a valid proposal	VERIFIED (shape)
§2 invocation budget: hard exact integer, never a forecast	No budget field in the live envelope	PROPOSED
§2 I-HEAT: dissipation rate within thermal envelope, metered	No meter, no envelope check in any live route	PROPOSED
§3 depth-gated allotment, tiers	No depth-tier allotment function in the live system	PROPOSED
§3 Boltzmann branch weights $w_k \propto e^{-\beta F_k}$	No Boltzmann weights in the live ranking; ranking is confidence descending, identifier ascending	PROPOSED
§3 golden-spiral widening on readiness stall	No widening step in the live loop; loop stops on Clarify	PROPOSED

Spec section	Current system component	Status
§3 anti-gaming seam (I-SB-1)	Closest analogue: exact identity checks and receipt-only persistence; no forecast-hiding grep gate has been run	PROPOSED
§4 deterministic normalization, FoldAttention	No FoldAttention normalization in the live backend; deterministic ranking and fixed-point confidence are the live determinism	PROPOSED
§5 seam: telemetry counts one way, budgets and scope the other	Telemetry is observation only in the live design (HyperDX rows record, they do not decide); no Meta-plane forecast seam is live	PROPOSED
§5 fast revoke on guardrail trip	Loop stops on Clarify/Reject with no call and no confidence change — the shape of a fast revoke, but no revocation drill has been run	PROPOSED
§5 slow reallocate over k ticks	No reallocation proof has been run; confidence moves only on verified outcomes, which is the ranking input a reallocation would read	PROPOSED
§6 role phase: TPPE step	TPPE is a reported kernel path in the specification's sources, not a live backend component in this mapping	REPORTED
§6 role carrier: harmonic MACs	Reported kernel path, not a live backend component in this mapping	REPORTED
§6 role reduce: TBIK tree	Reported kernel path, not a live backend component in this mapping	REPORTED
§6 role route: AttentionEconomics allocator	The live automaton loop is the closest analogue (gate, rank, receipt), but it allocates admission, not integer energy	PROPOSED
§6 role audit: invariant checker, fail-closed	Governed execution, exact-identity attestation, and the verified-outcome refusal are the live audit, and they fail closed	VERIFIED
§6 audit plus I-HEAT	No heat check in the live audit	PROPOSED
§7 hold: $\leq$12 branches, zero tokens while holding	Held proposal set under the automaton loop; no 12-branch cap and no Boltzmann holding rule is live	PROPOSED
§7 vary: counterfactual iteration, holonomy	No counterfactual iteration in the live loop	PROPOSED
§7 select: minimum-F branch, compliance permitted	Live select is gate Admit plus deterministic re-rank; no F is computed	PROPOSED
§7 distill: carved 1-cell arc, $O(1)$ next pass	Closest analogue: atom-intake gate admitting exactly one atom per passed run; no carved-arc distillation step is live	PROPOSED

Spec section	Current system component	Status
§7 re-gate: receipt checked against every invariant, including own cost	Receipts are persisted and confidence persists only on verified moves; no own-cost gate on a recursion turn is live	PROPOSED
§7 monotonicity: expected R non-decreasing, else STALLED	No monotonicity measurement; for the ratio form it inherits the refutation (monotone in R is not monotone in F); for R' monotonicity is definitional, measurement still unrun	PROPOSED
§7 dreamstate expansion on stall	No dream-state consolidation runs in the live system	PROPOSED
§8 spirals, not corkscrews: radius expansion gate	No radius-expansion gate in the live backend; a kernel gate, not a backend gate	PROPOSED
§9 gate 1: integer-only audit	Fixed-point confidence arithmetic is live; no kernel-source grep gate has been run for this system	PROPOSED
§9 gate 2: CORDIC correctness vs oracles, fuzz	No CORDIC fuzz gate has been run for this system	PROPOSED
§9 gate 3: no-corkscrew, variance $\geq 1000\times$ noise floor	No such measurement for this system	PROPOSED
§9 gate 4: reward integrity (A/B vs F-only optimizer)	Refuted as an equivalence for the ratio form; superseded for the objective by the $R' = -F$ proof (Mapping (4b)), which is definitional, not an A/B result; live metering still unrun	REFUTED (ratio) / FIXED (objective) / UNRUN (live)
§9 gate 5: seam audit, zero forecast fields	No spec grep gate has been run	PROPOSED
§9 gate 6: revocation drill, one tick, fail-closed	No drill has been run	PROPOSED
§9 gate 7: reallocation proof, dissolution path only	No proof has been run	PROPOSED
§10 register: preregister constants, fuzz harness, ablation	No register entries have been applied for this system	PROPOSED
Governed mind tools, disposable mind	Governed call failed closed: “mind child authority or contract attestation mismatch”; contract not verified, auth not reaching NS/DB	BLOCKED

Spec section	Current system component	Status
Native multi-token executor in the backend	Exists in code (`NativeMtpExecutor`), used by no HTTP route	PROBE-ONLY
Glimmer forward smoke	Input [954, 7963, 323, 11698, 373] to output [13796, 26, 589, 7963, 323], digest and receipt as in Mapping (6)	VERIFIED
Math gate executed, not claimed	Claimed 56, executed 56.0, ADMIT, receipt as in Mapping (6)	VERIFIED

Mapping (4): The November 2 counterexample, in plain arithmetic — REFUTED

Statement. Maximizing $R = Q/(H + c \cdot N)$ does not necessarily minimize $F =$ complexity $-$ accuracy, where complexity is $H + c \cdot N$ and accuracy is Q.

Working.

1. Define $C = H + c \cdot N$, so that $R = Q/C$ and $F = C - Q$, by the specification's own definitions.
2. Action A: $(Q, C) = (0.9, 3)$. Then $R_A = 0.9/3 = 0.30$, and $F_A = 3 - 0.9 = 2.1$.
3. Action B: $(Q, C) = (0.2, 1)$. Then $R_B = 0.2/1 = 0.20$, and $F_B = 1 - 0.2 = 0.8$.
4. Compare rewards: $0.30 > 0.20$, so A is the reward-maximizing action. Compare free energies: $2.1 > 0.8$, so A has the higher free energy, and B has the lower free energy.
5. Therefore the reward-maximizing action is not the free-energy-minimizing action, on these numbers. The claim "maximizing R also minimizes F" is false as stated. One counterexample suffices to refute a universal claim, and this is that counterexample.
6. Why it diverges, in one line: a ratio rewards high quality per unit cost, even at high absolute cost, while a difference rewards low absolute cost minus quality. A high-quality, high-cost action can win the ratio and lose the difference, which is exactly what Action A does.
7. What survives the refutation: R as a separate objective, to be tested on its own, with a declared quality floor, a resource budget, and measured-cost calibration. What does not survive: R as a stand-in for F, and any alignment guarantee argued from Landauer's lower bound alone.

Frozen inputs. The counterexample $(Q, C) = (0.9, 3)$ versus $(0.2, 1)$, from the specification's correction entry dated November 2, 2026. The correction entry also cites an exact-integer counterexample and a bounded enumeration in `results/forward-recall-math/report.tex` and `scripts/verify.rs`; those files are not reproduced here, and no enumeration result beyond the stated counterexample is claimed in this paper.

What would settle it. Nothing further is needed for the refutation; it is arithmetic. What remains is the narrower test: whether R, as a separate objective with a quality floor and a budget, ranks actions usefully on a fixed corpus. That test is unrun.

Mapping (4b): The November 4 fix — $R' = -F$ — FIXED AT THE OBJECTIVE DEFINITION

Statement. Replace the ratio reward with the linear reward $R' = Q - C$, where $C = H + c \cdot N$, so that $R' = -F$. Then maximizing R' is minimizing F, by definition, for every pair of actions. The old ratio Q/C is demoted to a reporting diagnostic only, never the optimizer's objective.

Working.

1. Define $C = H + c \cdot N$ and $F = C - Q$, by the specification's own definitions. Define $R' = Q - C = -F$.
2. For any two actions A and B: $R'(A) > R'(B)$ if and only if $-F(A) > -F(B)$, if and only if $F(A) < F(B)$. The same chain runs in reverse. No counterexample is possible, because the claim is an identity with a sign reversed, not an empirical correlation.
3. Re-check the old counterexample. Action A: $(Q, C) = (0.9, 3)$, so $R' = 0.9 - 3 = -2.1$ and $F = 2.1$. Action B: $(Q, C) = (0.2, 1)$, so $R' = 0.2 - 1 = -0.8$ and $F = 0.8$. Action B now wins reward ($-0.8 > -2.1$) and wins free energy ($0.8 < 2.1$). Alignment is restored.
4. Why the ratio diverged, and why the difference cannot: a ratio rewards high quality per unit cost even at high absolute cost, while a difference rewards low absolute cost minus quality. Changing the form from a quotient to a difference removes the divergence at its root.
5. Gates, which stay and are not the proof: quality floor $Q \geq \Omega_{\min} = 0.82$; budget cap $C \leq E_{\text{alloc}}$ for that invocation; verified outcome only; fail-closed with a receipt when no action is admissible. Action B above, with $Q = 0.2$, would also have been inadmissible on the floor alone. The floor is a guardrail, not the alignment proof.
6. What is unchanged, and what is not yet live: the live automaton confidence rule is unchanged (verified success $c' = c + (1 - c) \cdot 0.1$, verified failure $c' = c \cdot 0.9$, unverified outcomes refused), and it must not be rewritten to use H or Q until both are actually measured. R' is the declared objective, not a live reward signal, because H (powermetrics on Apple Silicon, RAPL on Linux) and judge-graded Q are not yet metered per invocation, and c is not yet preregistered per substrate from measurement.

Frozen inputs. The counterexample $(Q, C) = (0.9, 3)$ versus $(0.2, 1)$, from the November 2 correction entry; the fix definition in `reward-free-energy-fix-2026-11-04.md`, dated November 4, 2026, status FIXED AT THE OBJECTIVE DEFINITION.

What would settle it, for live policy. Three wiring items, none yet present: H metered at the sink per invocation, not estimated; Q from a deterministic judge per task, receipted; c preregistered per substrate, measured not assumed. Until those three exist, $R' = -F$ is a fixed definition awaiting instrumentation, and any claim that it is already live policy would repeat the ratio bug in a new form.

Mapping (5): The blocked tool leg, stated as a result — BLOCKED

Statement. A governed mind tool call for a disposable mind, on November 4, 2026, failed closed and did not run.

Working.

1. The call was POST `/api/harness/mcp/execute`, tool `mind.db_status`, authority endpoint `http://127.0.0.1:8000`, namespace `ares_disposable`, database `mind_system_run`.
2. The backend attests the machine-readable child before any tool runs. Attestation requires status ok, mode On, contract verified, and exact endpoint, namespace, and database.
3. The result was the error "mind child authority or contract attestation mismatch". The disposable database as configured did not attest: its contract was not verified, and its authentication did not reach that namespace and database.
4. The call did not run, was not parsed into a result, and was not silently dropped. It failed closed with a why-receipt.
5. Consequence, stated in the system task itself: the disposable mind must attest, by a seeded contract or working authentication for that namespace and database, before any governed mind tool, or any mission that depends on one, can run. No mission rerun for a green receipt is valid until that attestation passes, because it uses the same attest step and would fail the same way.

Frozen inputs. The system-task baseline of November 4, 2026, tool leg, verbatim error and authority values above.

What would settle it. A seeded contract or working authentication for the disposable namespace and database, followed by a governed tool call that attests and runs. Until then, the tool leg is BLOCKED, and every brigade claim that depends on governed mind tools inherits that block.

Mapping (6): The measured numbers, with working shown — VERIFIED, with scope

Statement. The draft acceptance, decode rates, Glimmer forward, math-gate, and swarm numbers in the front follow from their receipts by the arithmetic shown.

Working.

1. Acceptance is $1320/1368$. Divide: $1320 \div 1368 = 0.964912\ldots$, reported as 0.9649. Failures and unfinished are both 0 in the same bench, so the acceptance fraction is not inflated by dropping failed cells. Bench scope: 2000 exact-reference closed-loop tokens, short-context HTTP controls, not production or broad model qualification.
2. Bench throughput is 38.664 tokens per second on lanes 1 and 38.896 on lanes 3, for the same 2000 tokens. System-task warm decode is $32/803$ ms $= 39.8$ tokens per second on warm run 1 and $32/792$ ms $= 40.4$ tokens per second on warm run 2, each for 32 tokens after first token. The headline "40 tokens per second, warm" is the rounded warm-decode range, not the bench rate, and the two are never averaged together in this paper.
3. Warm time to first token is 262–425 milliseconds across the two warm system-task runs. Cold time to first token is 18419 milliseconds on a first-after-idle request. The cold number is reported separately, as its own cost, not folded into the warm number.
4. Native multi-token server policy, from the live information frame: execution Mtp, draft width 4, stream capacity 1, max proposers 1, work limit 3, initial mass 2147483647, admission mass 268435455, reward mass 536870911, qualification trials 64, probe every epochs 2, layer group 40. Seat for the multi-token server is the Qwen seat, not the Glimmer seat.

5. Glimmer forward, input [954, 7963, 323, 11698, 373], output [13796, 26, 589, 7963, 323], logits digest `beeaf3041a91bb4b625b18725b2972111d063863f24cc6158c63edb8783e0b2d`, receipt `b03e9753bf9528c84c4d2a5b5578abea2a58682973b28fb5ff5120ab7561ca35`, seat `1885e3a3adc075c315fc4e892abd513e23bccebc3ce0b049a38facb443e7429e`, 6.9 seconds alone and 30.5 seconds with the multi-token server also resident.

6. Math gate, claimed 56, executed 56.0, receipt `e895d7ce0bd82cb397baae6965d52969e7b0bdb86605658f26480d08c670b6e1`, calibration count 1, confidence raw 1374389534, which is $1374389534/2^{31} = 0.64$ in Q32.31.

7. Swarm: 10 runtime minds, 9 effective, 1 failed (Pax lock: premises right, conclusion wrong, as stated in the front). Task effectiveness is $9/10 = 0.9$, and it is not an acceptance rate. Distractors were ignored in all 10 outputs. Minds were prompt-seeded through the multi-token server, not Surreal-attested minds.

Frozen inputs. Native multi-token legit bench, November 4, 2026; system-task baseline, November 4, 2026; swarm at small scale, November 4, 2026; Glimmer forward smoke, November 4, 2026; automaton loop fix report, November 4, 2026.

What would settle it. Nothing further for these numbers; they are receipts. What remains is the next ablation, against the headline number in the front, plus the attestation fix, which would unlock a tool leg for the system task.

What was dropped for lack of verification

No measured number is reported in this paper that could not be verified against live code or a stated receipt. In particular: no heat-metered reward, no energy budget, no Boltzmann branch weight, no free-energy value, no dream-state consolidation result, no revocation or reallocation result, no kernel-gate result, and no Cloudflare-hosted Glimmer result are included, because none exists to cite. Each is listed above as PROPOSED, UNRUN, BLOCKED, or REFUTED instead. No external paper is cited with a page, volume, or DOI, because no external claim in this paper depends on one.

Sources

Live code, specifications, and receipts only. No external citation with page, volume, or DOI is used.

- Specification mapped: `AttentionEconomicsAndActiveInference-Spec.md`, dated 2026-10-27, status PROPOSED, with correction entry dated 2026-11-02 refuting the ratio reward-free-energy equivalence, and fix definition `reward-free-energy-fix-2026-11-04.md`, dated 2026-11-04, status FIXED AT THE OBJECTIVE DEFINITION ($R' = Q-(H+c\cdot N) = -F$; ratio retained only as a diagnostic).
- `ares-backend-loop/src/harness/automaton.rs` — `ProposerFamily`, `ProposerEnvelope`, `AuditSignals`, `StrategyState`, `Verdict` (`Admit` / `Reject` / `Clarify`), `transition`, `update_from_outcome`, `rerank_strategies`, governed execution route.
- Automaton loop fix report, November 4, 2026 — `rerank_strategies`, `run_automaton_loop`, verified-outcome-only confidence persistence, loop pressure test.

- Native multi-token server, draft policy fields and live information frame — execution Mtp, width 4, masses and trials in Mapping (6).
- Native executor: `ares-backend-loop/src/harness/native_mtp.rs` — exists in code, used by no HTTP route, probe-only.
- Math gate: `ares-backend-loop/src/harness/math_gate.rs` — executed value, receipt, confidence.
- Measured receipts, November 4, 2026: native-mtp-legit-20261104 bench; system-task-baseline-2026-11-04; swarm-10-runtime-minds-2026-11-04; glimmer-forward-smoke-2026-11-04.
- Logos: the two Atoma Labs atom marks supplied November 4, 2026, cropped as a diptych on the cover; the old A-ring logo is not used.
- House format: slate matte grey header band, as in the two prior papers of November 4, 2026, in place of the navy band.

No external paper is cited with a page, volume, or DOI in this preprint, because no external claim in it depends on a citation that was not checked against its source in this build.